



Simultaneous Calibration of Noise Covariance and Kinematics for Legged State Estimation via Bi-level Optimization

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We turn hand-tuned estimator parameters into optimization variables by differentiating through a full-information estimator.

Problem & Key Idea

Motivation:

Legged estimators fuse **IMU**, **joint encoder**, and **contact measurements**. Uncertain **covariances** make hand tuning slow and sub-optimal, while **kinematic errors** can systematically bias the estimator.

Method:

We jointly calibrate covariances and kinematics via **bi-level optimization**. Lower level estimates the trajectory; upper level minimizes estimation error.

Outcome:

Calibration **improves** velocity and attitude estimates while **reducing drift**. Learned parameters **transfer** to new trajectories with **low** estimation error.

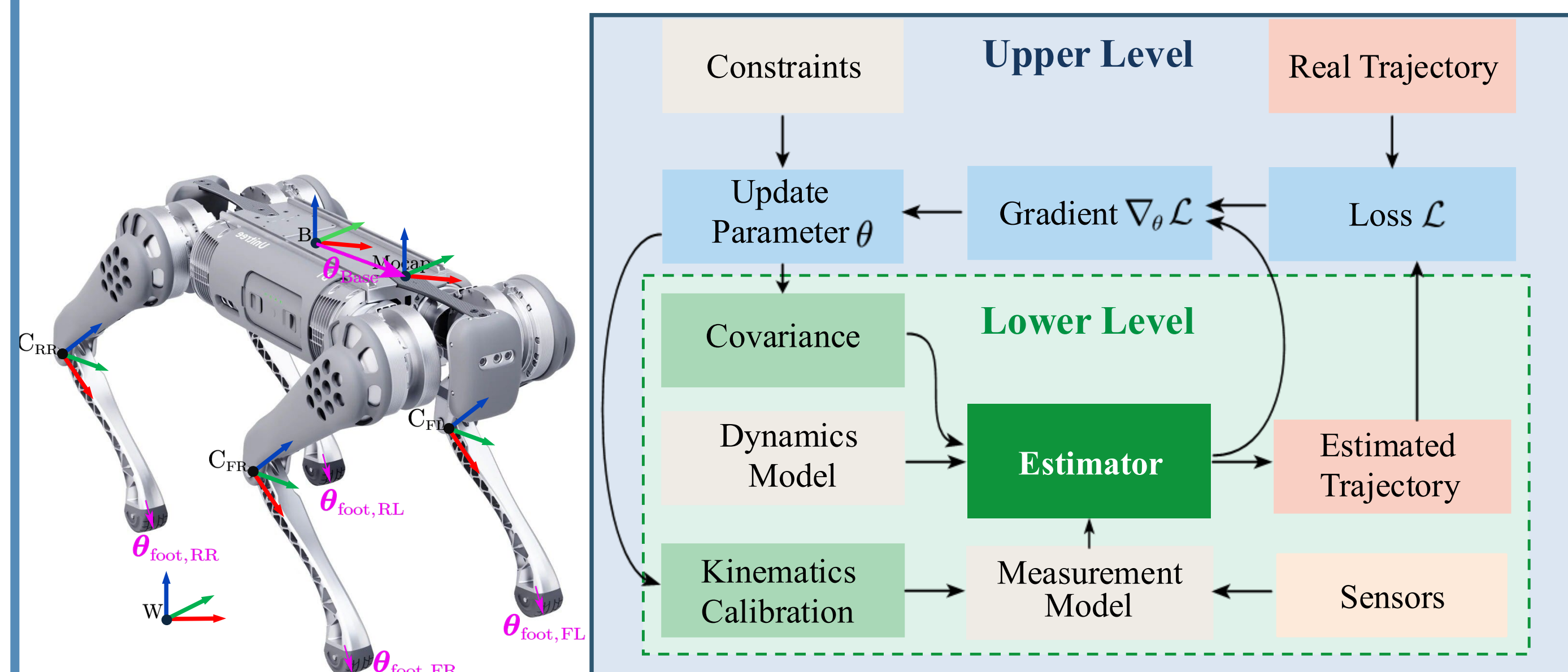


Figure 1. Calibration updates covariances and kinematic offsets from trajectory-level error.

Lower Level: Full Information Estimation

Given calibration parameters θ , the lower level estimates the entire state trajectory by solving a full horizon Maximum a Posteriori problem:

$$\min_{\mathbf{x}_{0:T}} \mathcal{J} = \|\mathbf{x}_0 - \tilde{\mathbf{x}}_0\|_{\mathbf{P}_{0-1}}^2 + \sum_{k=0}^{T-1} \|\mathbf{w}_k\|_{\mathbf{Q}_k}^2 + \sum_{k=0}^T \|\mathbf{n}_k\|_{\mathbf{R}_k}^2$$

s.t.
 IMU Integrated Process Constraint
 Joint Encoder Measurement Constraint
 Contact Physical Constraint

θ enters the estimator by changing residual weights and joint kinematics

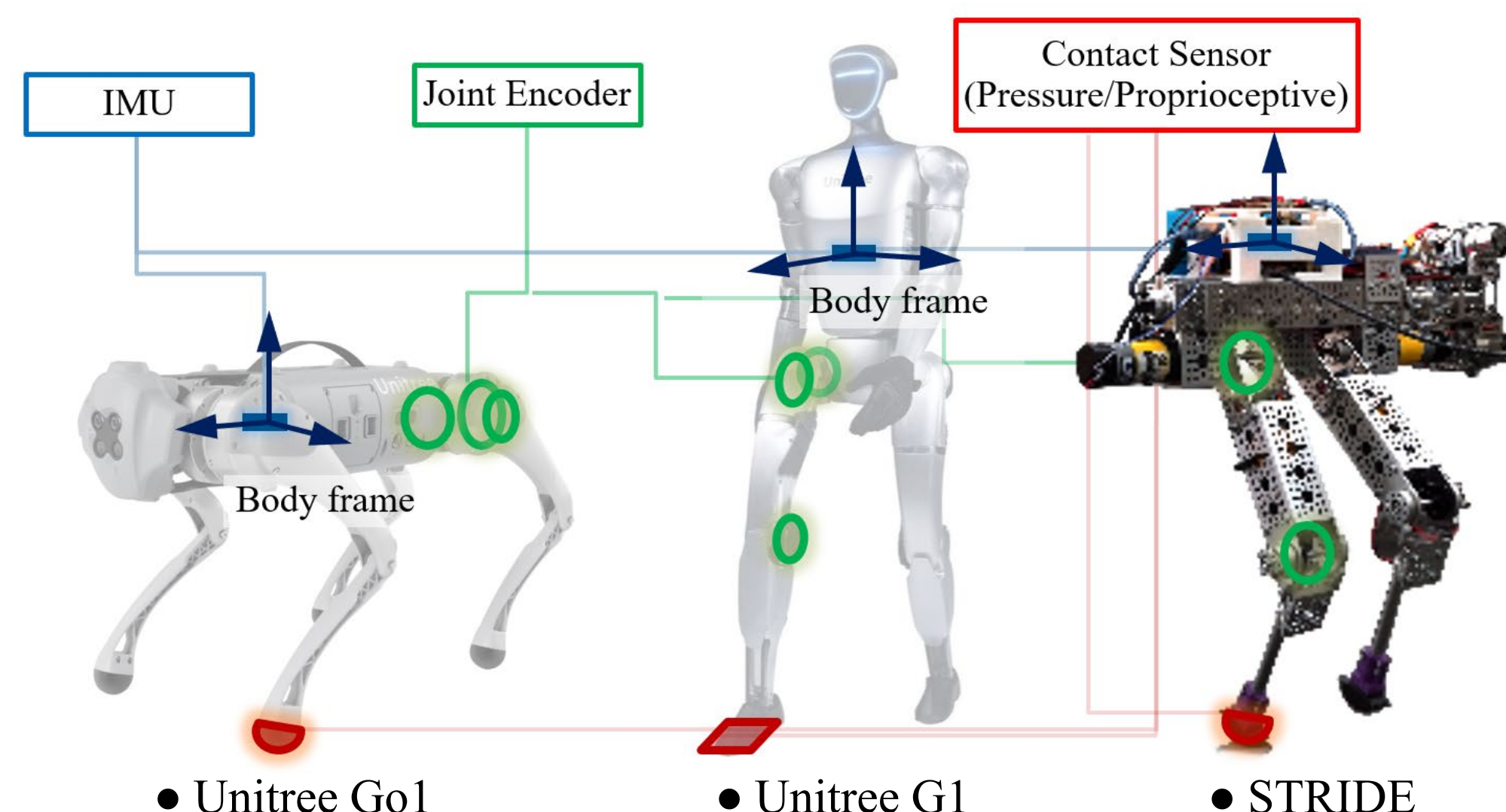


Figure 2. common sensing capabilities on legged robots.

Bi-level Optimization Calibration

Bi-level Optimization Framework

We calibrate θ by minimizing a trajectory-matching loss on manifold: $\mathcal{L}$ measures the error between the estimated trajectory and ground truth.

$$\min_{\theta} \mathcal{L}(\hat{\mathbf{x}}(\theta), \mathbf{x}_{GT}, \theta),$$

$$\text{s.t. } \theta \in \mathcal{C}, \quad \hat{\mathbf{x}}(\theta) = \operatorname{argmin}_{\mathbf{x}} \mathcal{J}(\mathbf{x}, \theta).$$

- **Upper level:** choose calibration parameters.
- **Lower level:** solve the full-information estimator.

θ includes covariance parameters for $\mathbf{P}_0, \mathbf{Q}_k, \mathbf{R}_k$ and kinematic offsets θ_{kin} .

Estimator Differentiation + Frank-Wolfe Optimization

- **Differentiate through the estimator** via the **Implicit Function Theorem** applied to the KKT condition of the estimator:

$$\mathcal{F}(\mathbf{x}^*, \theta) = 0$$

$$\left(\frac{\partial \mathcal{F}}{\partial \mathbf{x}} \right) \frac{\partial \mathbf{x}^*}{\partial \theta} = - \frac{\partial \mathcal{F}}{\partial \theta}, \quad \nabla_{\theta} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}^*} \frac{\partial \mathbf{x}^*}{\partial \theta} + \frac{\partial \mathcal{L}}{\partial \theta}.$$

- Frank-Wolfe step for θ updates:

$$s_t = \arg \min_{s \in \mathcal{C}} \langle s, \nabla_{\theta} \mathcal{L}(\theta_t) \rangle, \quad \theta_{t+1} = (1 - \alpha_t) \theta_t + \alpha_t s_t.$$

With a convex objective and Linear Matrix Inequality (LMI), each Frank-Wolfe subproblem is a semidefinite program (SDP).

Results: Simulation (Noise + Calf Bias)

Calibration recovers the kinematic bias and improves the estimation accuracy.

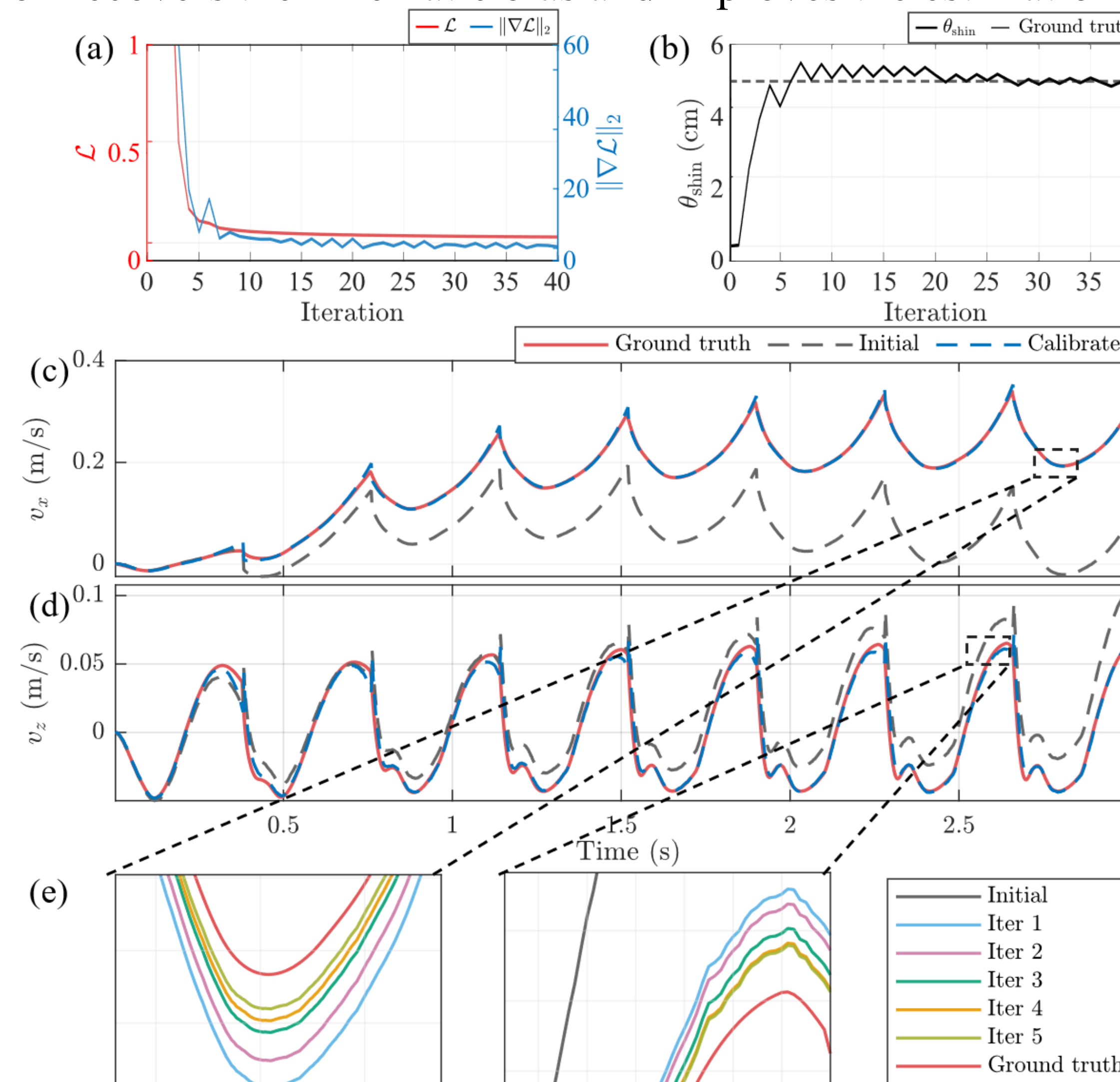


Figure 3. Simulation calibration: convergence, calf-bias recovery, and velocity tracking.

Results: Quadrupeds Real-World Calibration

Parameters calibrated on Unitree B1 hardware generalize to a new walking trajectory, improving estimation accuracies.

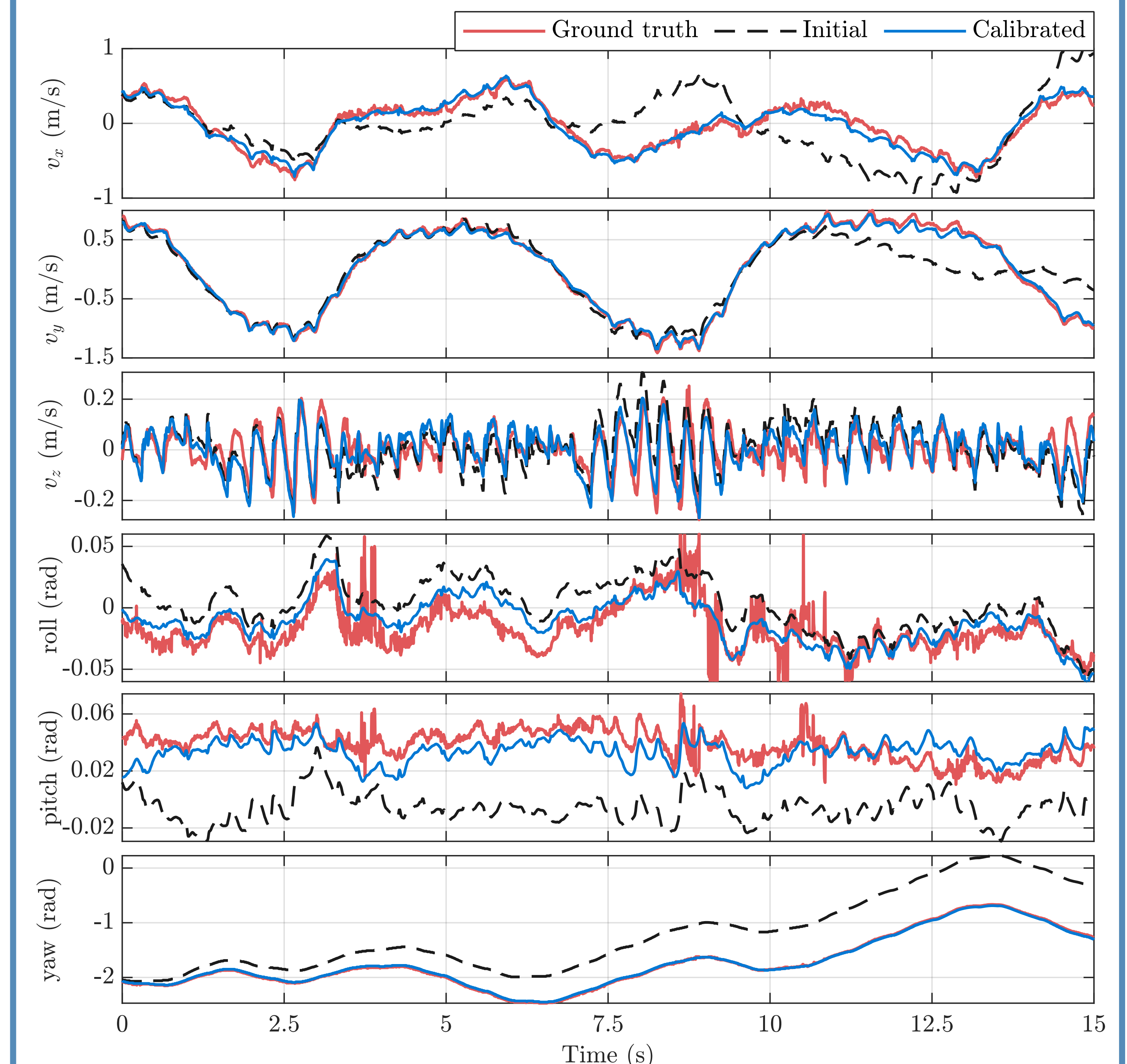


Figure 4. B1 hardware experiment: calibrated covariances and kinematics reduce drift in body velocity and attitude.

Table 1. RMSEs of B1 hardware estimates before calibration, after calibration, and on a new evaluation trajectory

Estimation Metric	Before Calib.	After Calib.	New Eval.
RMSE _v [m/s]	0.2658	0.0610	0.0706
RMSE _{Euler} [rad]	0.3457	0.0151	0.0514

Calibration reduces **velocity RMSE by 77.1%** and **attitude RMSE by 95.6%**, while maintaining low error on unseen data.

Future Works

- **Identifiability:** analyze identifiability of covariance and kinematic calibration parameters.
- **Generalization:** extend estimator calibration to legged, aerial, and wheeled robots with exteroceptive sensing.
- **Faster optimization:** enable online, real-time calibration from onboard IMU, encoder, and contact measurements.

